Misspecification results summary

DR vs NLL under controlled residual dials
Full-catalog runs only

OPC project

3 November 2026 (Asia/Jerusalem)

Abstract

Locked confirmatory numbers for the misspecification dial study on **original full catalogs** (MovieLens-1M, Anime, Myket; KuaiRec/KuaiRand Phase-0a status noted). Primary estimand:

$$D(\gamma) = V(\pi_\gamma^{\mathrm{OPC/DR}}) - V(\pi_\gamma^{\mathrm{NLL}}), \qquad \mathrm{DiD} = D(\gamma^\star) - D(0),$$

with exact-greedy CTR in percentage points. We report absolute CTR levels and DiDs for residual family $R^\star$, hybrid NLL→DR, latent SoftAND*/SoftThresh, and CausE/BLOB references. Catalog-geometry interventions (mlsize / amp / damp / head-match) are **out of scope** for this note.

1 Setup (reminder)

- Truth logit: affine-bilinear clean score $+\,\gamma\,\alpha\,\phi$ with $\mathrm{sd}(\phi) = 1$; logger temperature frozen at $\gamma = 0$; reference CTR 5%.
- Primary family $R^\star$: even map on a second random projection (s=1, m=4, ψ= tanh).
- Phase-1 learners: matched linear capacity; methods OPC/DR, NLL, optional `hybrid_seq_f0.5`, n=10^5 train / $2{\cdot}10^4$ val, 20 Optuna trials.
- Bias cell: combined-high. Embeddings: `BPR/embeddings` (full).

2 $R^\star$ on MovieLens (identifying)

Locked $\gamma^\star$=0.5, seeds 300–309 (n=10).

Table 1: Exact-greedy CTR (%) on ML-1M under $R^\star$.

method	γ=0	γ=0.5	Δ level	—
OPC/DR	16.99	18.42	+1.43	
NLL	18.06	17.96	−0.09	
D=$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	−1.07	+0.46	DiD=+**1.53** pt	

DiD= + 1.53 pt, 95% CI [+0.97, +2.09], signs 10/10. Go checks pass. NC1 control DiD ≈ +0.20 (within null band).

3 Hybrid NLL→DR on MovieLens

Sequential schedule $f^\star$=0.5 (half epochs NLL, then raw DR). Fresh 20-seed run (`nlres_h1_rstar_f05`, seeds 300–319).
Hybrid keeps near-NLL level at γ=0 and leads absolute CTR at $\gamma^\star$.

Table 2: ML hybrid levels and DiDs (n=20, $\gamma^\star$=0.5).

method	γ=0	γ=0.5	H−NLL DiD	OPC−NLL DiD
NLL	18.27	18.63	—	—
OPC	17.17	18.88	—	+1.35 [+0.88, +1.83] (19/20)
hybrid f=0.5	18.07	**19.62**	**+1.19** [+0.59, +1.79] (17/20)	—

4 Phase-2 references: CausE / BLOB (ML only)

Matched-capacity OPC-mlp64 vs CausE-cap; published BLOB-L10-nq and CausE-linear as refs. Seeds 300–309.

Table 3: Phase-2 absolute greedy CTR (%) and DiDs vs OPC-mlp.

γ	OPC-mlp	CausE-C-mlp	CausE-C-lin	BLOB-L10
0	17.54	19.00	18.90	19.28
0.5	19.55	18.68	19.04	**20.09**

- OPC−CausE-C-mlp DiD= + 2.32 [+1.11, +3.54], 9/10.
- OPC−BLOB DiD= + 1.20 [+0.54, +1.85], 8/10 — OPC gains relatively, but BLOB still wins **level** at $\gamma^\star$.

5 Latent shift families (ML)

Protocol `misspec-latent-alt-v1.0`. SoftAND* locked at $\gamma^\star$=0.5 by Phase-0a; SoftThresh $\tau \in \{1, 1.5, 2\}$ as runners-up. Phase-1 linear DiD:

Table 4: Latent Phase-1 OPC−NLL DiD on ML (n=10).

family	$\gamma^\star$	DiD (pt)	95% CI	signs
SoftAND*	0.5	+0.09	[−0.11, +0.29]	7/10
SoftThresh τ=1	0.5	−0.20	[−1.19, +0.79]	4/10
SoftThresh τ=1.5	0.5	+0.02	[−0.80, +0.84]	4/10
SoftThresh τ=2	0.25	+0.02	[−0.51, +0.55]	5/10
SoftAND*	100	+3.76	[+2.07, +5.46]	10/10
SoftThresh τ=1	100	+4.64	[+2.84, +6.45]	10/10
SoftThresh τ=2	100	+7.17	[+5.24, +9.10]	10/10

Reading. SoftAND/SoftThresh open oracle gaps at moderate γ but yield **null** linear DiD at the locked $\gamma^\star$; DR wins only under extreme $\gamma \to 100$ stress. Spine remains $R^\star$.

6 Multi-dataset $R^\star$ (full catalogs)

Phase-0a passers on full embeddings: **anime** ($\gamma^\star$=0.6), **myket** ($\gamma^\star$=0.5). Kuairand / KuaiRec: Phase-0a **NO-GO** (G2/G4 sandwich); full-catalog Phase-1 with hybrid+CausE/BLOB is running on CPU and not yet complete—omitted here.

Confirmatory seeds extended to n=20 (310–329) to stabilize hybrid CIs.

Table 5: Full-catalog confirmatory CTR (%) at $\gamma^\star$ and DiDs (n=20).

dataset	NLL / OPC / hybrid	OPC−NLL DiD	H−NLL DiD
Anime ($\gamma^\star$=0.6)	16.65 / 16.19 / **17.37**	+0.10 [−0.26, +0.46] (8/20)	+**0.50** [+0.08, +0.92] (16/20)
Myket ($\gamma^\star$=0.5)	20.60 / 19.74 / **21.37**	+0.55 [−0.20, +1.31] (11/20)	+**0.69** [+0.16, +1.22] (14/20)

Reading. On anime/myket, pure OPC DiD stays weak/null; **hybrid** clears a positive H−NLL DiD once n=20 (first-10 CIs had crossed zero). Absolute level: hybrid > NLL > OPC at $\gamma^\star$.

7 Takeaways

1. $R^\star$ on ML: confirmatory DR advantage DiD= + 1.53 pt is solid.
2. Hybrid f=0.5 is the practical interpolant: best absolute CTR under mid misspec on ML, and the only clear DiD win on anime/myket.
3. CausE is overtaken by OPC-mlp in DiD under $R^\star$; BLOB keeps a level edge.
4. Latent SoftAND/SoftThresh: oracle-active, learner-null at operational γ; stress γ=100 recovers large DiDs.
5. Kuai full-catalog learner results: pending (Phase-0a failed gates; confirmatory still in flight).

Artifacts: `artifacts/full_study/run_nlres_*` tags cited above; snapshot `results_snapshot.json`. This PDF intentionally excludes catalog-shrink / geometry-repair arms.